MathWorks TECHSOURCE Anata



Learn to hack:

Matlab Edition



Welcome, Participants!

Agenda

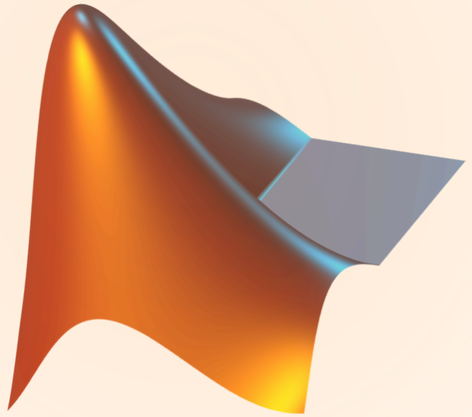
01 INTRODUCTION OF PROBLEM STATEMENT

02 SUBMISSION GUIDELINES

03 JUDGING CRITERIA

04 GETTING STARTED

05 HACKATHON TIMELINE



Problem Statement

PREDICTING BUILDING ENERGY CONSUMPTION

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- Develop a machine learning model that predicts building energy consumption factors such as:
  - Weather, Building Characteristics, etc
- Buildings are major contributors to global energy use and carbon emissions
- ML model to make accurate predictions which can help organisations to implement targeted strategies to reduce energy waste
  - Retrofitting high-use buildings to be more energy efficient
- Work on a real dataset on a current global issue and enhance your technical skills!

DATASET WILL BE SENT IN THE TELEGRAM CHANNEL!! Ensure that atleast one team member is in the channel, if not approach us after this briefing!

EXAMPLE MODEL:

- We have provided a very basic example that can be used as a reference when developing your own solutions
- This can be found in the MATLAB Live Script called **‘Building Regression Models.mlx’** which explains each step of the process
- Use this as a starting point, then determine the next steps of the process, such as:
  - How you process your data
  - How you split the data for training, test, and validation, if at all
  - How you train your models, and which type(s) of models you wish to train

Submission Guidlines

- Projects must be submitted to Devpost
  - You will need to create a Devpost project in order to submit
  - Submissions are due by today (30/10/26), **4:00pm** sharp
- Submissions must be on GitHub, MATLAB Drive, Google Drive, File Exchange, or some similar online repository that can be made public
  - It must contain your team's project and a brief (<5 minutes) demo of this project
  - The demo may be a video, slide deck, essay, or any presentation that you believe best allows you to showcase your project
  - The link to the code must be present and must be accessible on the online repository
- Only one submission per team is required

Judging Criteria

Predicting Building Energy Consumption
Scoring Rubric

Solution Quality	Points (70)
Data Mastery: Makes informed choices about how they choose to clean, process, and separate their data.	20
Use of Machine Learning: Can justify their algorithm choices, iterates on their models, and can demonstrate that the model achieves some level of accuracy	20
Functionality: Error-free and runs without issues	10
MATLAB proficiency: Level of MATLAB knowledge demonstrated in executing the tasks	10
Readability: Clean, organized and easy to comprehend	10
Presenting Quality	**Points (30)**
Creativity: Interesting delivery methods that are informative	10
Quality: Technical execution of material and attention to detail	10
Concept: Engaging, coherent and appropriate	10

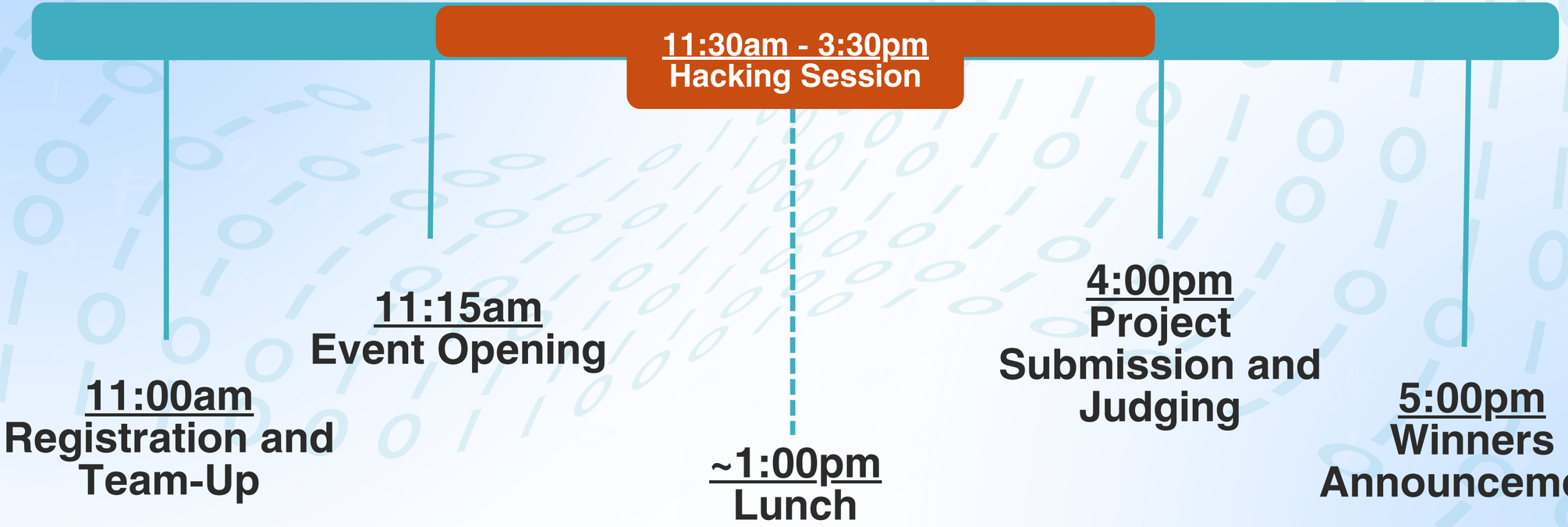
Getting Started

Start the devpost!
Assign one person to start the devpost project

Communicate!
Talk through the problem conceptually first before you start developing

Choose your role!
Once you have a plan, distribute accordingly so members will work on different parts of the problem

Hackathon Timeline



Judges Panel



Dr. Mukta Bansal

School of Chemistry, Chemical Engineering and Biotechnology



Asst. Prof. Liang Kaicheng

School of Chemistry, Chemical Engineering and Biotechnology



Dr. Wu

President & Co-Founder, ChemT Biotechnology

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Any Questions?

Feel free to approach any TA later for questions and queries!